

The MoCHII Atomic-Data Fitting Pipeline:

CHIANTI Cooling and n -level Fits

Abstract

MoCHII evaluates all metal-line atomic data from analytic fits; the raw CHIANTI database is never parsed at runtime. The Python pipeline under `tools/fitting/` is the auditable source of every coefficient in `data/atomic/`. This memo records the first production cycle (2026-07-11) for the ion set O II, O III, N II, S II, S III, Ne II, and Ne III: the Tier-1 cooling-function fits used inside the thermal iteration, the Tier-2 n -level data files (level energies, A -values, and Chebyshev fits of the effective collision strengths in Burgess–Tully scaled space) used for output-time line emissivities, and the verification of both against the CHIANTI v11.0.2 source data and against PyNeb. The fitted files reproduce direct CHIANTI evaluation to $\lesssim 0.5\%$ in the classic diagnostic ratios; the remaining differences from PyNeb (up to 6–10% for the density diagnostics near their critical densities) trace to PyNeb’s independent choice of collision strengths and A -values, not to fit error.

1. Overview: two tiers, one design rule

The metal-line design of `docs/PLAN.md` (section 8) requires that adding an ion be a data operation, not a code operation. The pipeline therefore produces, for each registered ion, self-contained coefficient files with provenance headers (CHIANTI version, population model, fit range, fit form, maximum fit error), split into two tiers:

- **Tier 1 — cooling fits (hot loop).** A smooth closed-form cooling function $\Lambda_{\text{ion}}(T)$, cheap and Newton-friendly inside the Stage-G2 thermal-balance iteration. Coronal (low-density) limit by construction.
- **Tier 2 — n -level data (diagnostics).** Level energies, statistical weights, radiative A -values, and fitted effective collision strengths $\Upsilon(T)$ feeding the generic n -level solver, evaluated only at output time on the converged state. This tier supplies the density-dependent emissivities and the diagnostic line ratios that Tier 1 cannot give.

The pipeline is seeded from EXHALE: `chianti_cooling.py` (copied from EXHALE/`cooling_data/`) reads the raw CHIANTI ASCII ion files (`.elvlc`, `.wgfa`, `.scups`) directly, with no ChiantiPy dependency, and applies the Burgess–Tully (1992) descaling of the stored scaled collision strengths. CHIANTI v11.0.2 lives at `~/RT_Codes/CHIANTI/dbase`.

2. The cooling function

For a transition $l \rightarrow u$ with energy gap ΔE_{lu} , the Maxwellian electron-impact excitation rate coefficient is

$$q_{lu}(T) = \frac{8.629 \times 10^{-6}}{g_l \sqrt{T}} \Upsilon_{lu}(T) e^{-\Delta E_{lu}/kT} \quad [\text{cm}^3 \text{s}^{-1}], \quad (1)$$

with $\Upsilon_{lu}(T)$ the effective collision strength obtained by descaling the `.scups` data. In the optically thin, low-density limit every collisional excitation is followed by a radiative decay, so the line cooling normalized by $n_e n_{\text{ion}}$ is

$$\Lambda_{\text{ion}}(T) = \sum_{l \rightarrow u} f_l(T) q_{lu}(T) \Delta E_{lu} \quad [\text{erg cm}^3 \text{s}^{-1}], \quad (2)$$

where $f_i(T)$ is the fractional population of the lower level. The pipeline uses the *ground-term* population model: the levels within 0.4 eV of ground (the ground-term fine structure, e.g. the $^3P_{0,1,2}$ block of O III) are Boltzmann-distributed and all excitations out of them cool. For the single $^4S_{3/2}$ ground level of O II and S II this reduces to the coronal limit.

3. Tier 1: cooling-function fits

Each $\Lambda_{\text{ion}}(T)$ curve is fitted over 10^3 – 10^5 K with the multi-exponential form used throughout EXHALE,

$$\Lambda_{\text{ion}}(T) = T^{-1/2} \sum_{i=1}^N A_i e^{-T_i/T}, \quad (3)$$

a Levenberg–Marquardt fit on $\log \Lambda$ with the T_i initialized from the level-energy ladder of each ion (fine structure / forbidden optical / UV blocks read off the `.elvlc` file). The fitter tries N , $N+1$, $N+2$ terms and keeps the best. One case needed care: Ne II cooling is a single fine-structure line ($12.8 \mu\text{m}$, $\Delta E/k \simeq 1123$ K) whose slow decay ($\Lambda \propto T^{-0.35}$, set by the T dependence of Υ) must be superposed from a ladder of T_i values before the far UV block at $\sim 3 \times 10^5$ K takes over; it converges with $N = 7$. Table 1 lists the fit quality; Figure 1 overlays the fits on the CHIANTI curves. The coefficients are written to `data/atomic/cooling_tier1_<ion>.txt`.

Table 1: Tier-1 fits, 10^3 – 10^5 K. The maximum relative error is against the direct CHIANTI evaluation of Eq. (2).

Ion	N	max error	output file
O II	5	0.07%	cooling_tier1_o_2.txt
O III	4	1.83%	cooling_tier1_o_3.txt
N II	6	0.14%	cooling_tier1_n_2.txt
S II	3	2.26%	cooling_tier1_s_2.txt
S III	6	0.46%	cooling_tier1_s_3.txt
Ne II	7	0.65%	cooling_tier1_ne_2.txt
Ne III	4	0.58%	cooling_tier1_ne_3.txt

A caveat carried in every file header: Eq. (3) is low-density-limit cooling. Transitions with low critical density (the fine-structure IR lines; [O II] $\lambda\lambda 3726, 3729$; [S II] $\lambda\lambda 6717, 6731$) saturate above n_{crit} , where the Tier-1 curve overestimates the cooling. If an application demands it, a fitted density-suppression factor can be added later; the emissivities themselves always come from Tier 2.

4. Tier 2: n -level data with fitted collision strengths

For each ion the lowest n levels are exported — enough for the optical/IR diagnostics (Table 2); extra levels are harmless to the solver. The file `data/atomic/nlevel_<ion>.txt` contains three blocks:

- NLEV: level index, statistical weight g , energy in cm^{-1} (observed values preferred, theoretical as fallback);
- NRAD: radiative rates A_{ul} from `.wgfa` (duplicate rows summed);

- NUPS: one line for each collisional transition — the Burgess–Tully type, scaling parameter C , transition energy in Ryd, the fitted domain $[s_{\text{lo}}, s_{\text{hi}}]$, a log flag, and the Chebyshev coefficients, followed by the recorded maximum fit error.

4.1 Burgess–Tully scaling and Chebyshev fits

CHIANTI stores Υ on the Burgess–Tully (1992) scaled temperature $s \in [0, 1]$. With $e_t \equiv kT/\Delta E$, the forward map is

$$s = \begin{cases} 1 - \frac{\ln C}{\ln(e_t + C)}, & \text{types 1, 4,} \\ \frac{e_t}{e_t + C}, & \text{types 2, 3, 5, 6,} \end{cases} \quad (4)$$

and the descaling back to Υ is

$$\Upsilon = y \ln(e_t + e) [1]; \quad y [2]; \quad \frac{y}{e_t + 1} [3]; \quad y \ln(e_t + C) [4]; \quad \frac{y}{e_t} [5]; \quad 10^y [6], \quad (5)$$

where $y(s)$ is the scaled value. Instead of the spline-through-table evaluation used by CHIANTI itself, the pipeline fits $y(s)$ with a Chebyshev series over the image of $T \in [10^3, 10^5]$ K intersected with the tabulated range,

$$u = \frac{2(s - s_{\text{lo}})}{s_{\text{hi}} - s_{\text{lo}}} - 1, \quad y = \sum_{k=0}^{n_c-1} c_k T_k(u), \quad (6)$$

clipping s into $[s_{\text{lo}}, s_{\text{hi}}]$ outside the fit domain. The degree grows from 2 until the descaled $\Upsilon(T)$ matches the direct spline evaluation to 0.2% (cap at 12). Weak intercombination transitions whose tables span decades (e.g. O III $^1D_2, ^1S_0 \rightarrow ^5S_2$) defeat a linear-space fit; for those the series stores $\log_{10} y$ instead, marked by the log flag (logf=1, apply 10^y before descaling). The Fortran `nlevel_mod` must evaluate exactly this recipe; it is replicated in `verify_nlevel_pyneb.py` as the reference implementation.

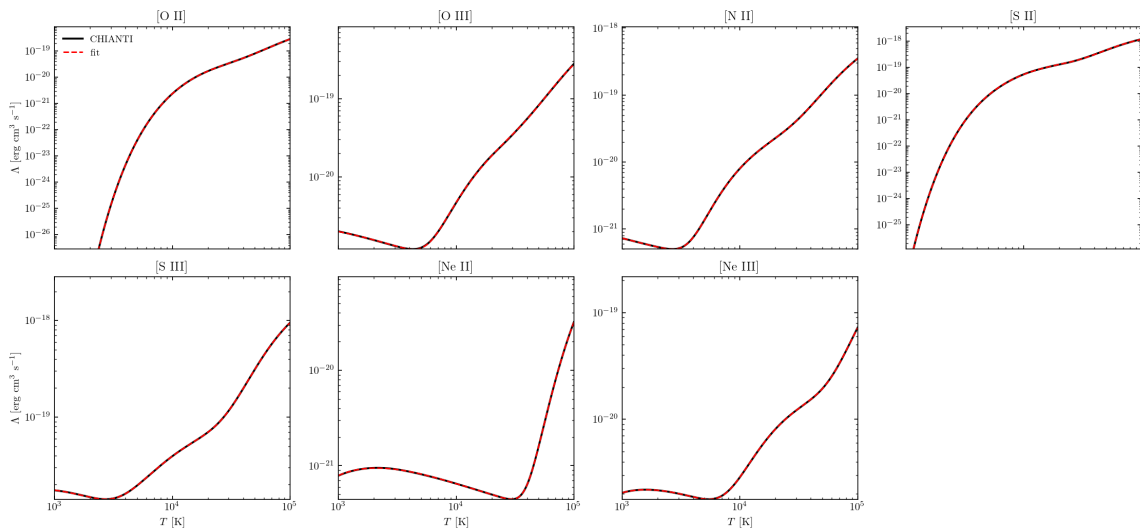


Figure 1: Tier-1 cooling fits (red dashed) over the direct CHIANTI curves (black), 10^3 – 10^5 K, ground-term population, low-density limit.

Table 2: Tier-2 exports. “worst Υ error” is the largest maximum relative error among the fitted transitions of that ion, measured on the descaled $\Upsilon(T)$ against the direct spline evaluation over 10^3 – 10^5 K.

Ion	levels	A-values	Υ fits	worst Υ error
O II	5	9	10	0.28%
O III	6	10	15	1.68%
N II	5	8	10	0.17%
S II	5	9	10	1.25%
S III	6	12	15	0.29%
Ne II	2	1	1	0.16%
Ne III	5	9	10	0.19%

The two entries above 1% have benign origins: the O III outlier is the physically negligible 1S_0 – 5S_2 intercombination transition ($\Upsilon \sim 10^{-4}$), and the S II errors reflect the coarse 10-point .scups tables of that ion — the deviation is from the interpolating spline, at the level of the resolution of the source data itself.

4.2 Statistical equilibrium and emissivities

Given (T_e, n_e) , the solver builds the rate matrix from the file contents alone: collisional rates from Eq. (1) with de-excitation by detailed balance, $q_{ul} = 8.629 \times 10^{-6} \Upsilon / (g_u \sqrt{T})$ and $q_{lu} = (g_u / g_l) q_{ul} e^{-\Delta E / kT}$, plus the radiative A_{ul} ; the level populations follow from $\dot{n}_i = 0$ with the closure $\sum_i n_i = 1$, and the emissivity of a line is $j_{ul} \propto n_u A_{ul} \Delta E_{ul}$.

5. Verification

5.1 Internal consistency: fitted files vs. direct CHIANTI

Running the same statistical-equilibrium solver twice — once reading the fitted `nlevel` files, once evaluating the raw CHIANTI splines and `.wgfa` rates directly — isolates the error contributed by the fits and the file format. Over the grid $T_e \in \{5, 10, 20\} \times 10^3$ K, $n_e \in \{10^2, 10^3, 10^4\} \text{ cm}^{-3}$:

Diagnostic ratio	worst fit/raw − 1
[O II] $\lambda 3726 / \lambda 3729$	0.05%
[S II] $\lambda 6717 / \lambda 6731$	0.52%

The fitted files are faithful to CHIANTI v11.0.2.

5.2 Cross-check against PyNeb

PyNeb (v1.1.31) carries its own atomic data — for these ions the collision strengths SSB14 (O III), T11 (N II), Kal09 (O II), TZ10 (S II) with A-values FFT04-SZ00, FFT04, Z82-WFD96, RGJ19 respectively — so this comparison is a cross-calibration against an independent reference, not a fit-residual test. Tables 3 and 4 give the classic temperature and density diagnostics.

Table 3: Temperature diagnostics vs. PyNeb.

T_e [K]	n_e [cm ⁻³]	[O III] $\lambda 5007/\lambda 4363$			[N II] $\lambda 6584/\lambda 5755$		
		MoCHII	PyNeb	diff	MoCHII	PyNeb	diff
5000	10 ²	4104	4153	-1.2%	829.6	828.6	+0.1%
5000	10 ³	4093	4133	-1.0%	818.3	817.2	+0.1%
5000	10 ⁴	3907	3943	-0.9%	624.7	623.9	+0.1%
10000	10 ²	152.0	153.4	-0.9%	66.97	66.93	+0.1%
10000	10 ³	151.7	152.8	-0.7%	65.75	65.64	+0.2%
10000	10 ⁴	146.0	147.0	-0.6%	52.23	52.12	+0.2%
15000	10 ²	50.97	51.30	-0.6%	29.06	29.08	-0.1%
15000	10 ³	50.77	51.10	-0.6%	28.53	28.59	-0.2%
15000	10 ⁴	49.08	49.43	-0.7%	23.26	23.32	-0.3%
20000	10 ²	29.82	29.78	+0.2%	19.23	19.23	+0.0%
20000	10 ³	29.58	29.67	-0.3%	18.92	18.92	-0.0%
20000	10 ⁴	28.68	28.81	-0.5%	15.71	15.71	+0.0%

Table 4: Density diagnostics vs. PyNeb.

T_e [K]	n_e [cm ⁻³]	[O II] $\lambda 3726/\lambda 3729$			[S II] $\lambda 6717/\lambda 6731$		
		MoCHII	PyNeb	diff	MoCHII	PyNeb	diff
5000	10 ²	0.8045	0.7641	+5.3%	1.298	1.333	-2.6%
5000	10 ³	1.436	1.383	+3.9%	0.8223	0.8396	-2.1%
5000	10 ⁴	2.364	2.491	-5.1%	0.5577	0.5052	+10.4%
10000	10 ²	0.7697	0.7411	+3.9%	1.309	1.351	-3.1%
10000	10 ³	1.260	1.241	+1.5%	0.8810	0.9139	-3.6%
10000	10 ⁴	2.267	2.397	-5.4%	0.5655	0.5203	+8.7%
15000	10 ²	0.7566	0.7341	+3.1%	1.320	1.349	-2.1%
15000	10 ³	1.180	1.179	+0.2%	0.9185	0.9520	-3.5%
15000	10 ⁴	2.209	2.345	-5.8%	0.5704	0.5294	+7.7%
20000	10 ²	0.7502	0.7319	+2.5%	1.343	1.347	-0.3%
20000	10 ³	1.136	1.144	-0.7%	0.9543	0.9786	-2.5%
20000	10 ⁴	2.170	2.311	-6.1%	0.5766	0.5373	+7.3%

The temperature diagnostics agree to 1.2% ([O III]) and 0.3% ([N II]). The density diagnostics drift to 6–10% at $n_e = 10^4 \text{ cm}^{-3}$, near and above the critical densities of the 2D levels, where the ratios become A -value-limited — and the two codes take their A -values from different compilations (CHIANTI v11 vs. Z82-WFD96 and RGJ19). Since the fitted files reproduce raw CHIANTI to $\leq 0.5\%$ (previous subsection), this spread measures the state of the atomic data, not the pipeline; it should be kept in mind as the floor of attainable agreement when the G2/G5 gates compare MoCHII line ratios with PyNeb-based references.

6. Adding an ion

The procedure is a data operation:

1. add one entry (element, stage, level-energy guesses / level count) to the IONS dictionary of `fit_cooling_tier1.py` and of `fit_nlevel_tier2.py`;
2. run both scripts; check the reported maximum fit errors and the overlay figure;
3. extend `verify_nlevel_pyneb.py` with the diagnostic ratios of the new ion where PyNeb has data, and record the comparison.

The suggested order of addition (cooling importance plus diagnostics, docs/PLAN.md) continues with C, Ar, and Fe.